

NEBULAOPS v22.3 FINAL

Architecture & Infrastructure Technical Specification

Professional architecture and runtime infrastructure document for NebulaOps v22.3 final release.

53

services

47

host mappings

8

volumes

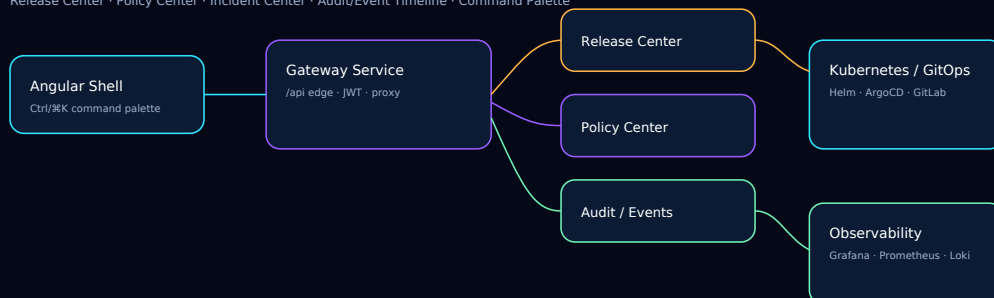
7

new ports

Professional release closure package - prepared for technical review, handover and local validation.

NebulaOps v22.3 · Operational Intelligence & Governance Layer

Release Center · Policy Center · Incident Center · Audit/Event Timeline · Command Palette



Reference figure - NebulaOps v22.3 platform topology.

1. Architecture scope

NebulaOps v22.3 is a local-first DevOps control plane composed of an Angular shell, deployable micro frontends, a Spring Boot gateway, domain services, data/messaging infrastructure, identity providers, observability components and GitOps/IaC tooling. The v22.3 release adds operational intelligence and governance capabilities on top of the existing architecture.

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v22.3 operational intelligence architecture overview.

2. Runtime layers

Layer	Components	Purpose
Client/UI	Angular shell, 13 micro frontends	Operator experience, navigation, dashboards, runtime actions
API edge	gateway-service	Single /api entry point, JWT validation, proxy and runtime orchestration
Governance	Release, Policy, Audit, Notification, AI Ops	Release gates, event timeline, incident analysis and notifications
Domain services	Auth, Task, File, Pipeline, DevSecOps, FinOps, GitOps, Env, Terraform, Observability	Business and operational APIs
Data plane	MongoDB, Redis, RabbitMQ	Persistence, cache and events
Observability	Grafana, Prometheus, Loki, Tempo, OTel	Metrics, logs, traces and dashboards
Delivery	Docker Compose, Kubernetes, Helm, ArgoCD, GitLab	Build, deployment and GitOps control plane

3. New v22.3 service exposure

Host port	Service	Purpose
4220	mfe-infra-hub	INFRA Hub MFE
4221	mfe-release-center	Release Center MFE
4222	mfe-policy-center	Policy Center MFE
4223	mfe-notification-center	Notification Center MFE
8098	release-orchestrator-service	Release API
8100	policy-governance-service	Policy API
8101	audit-service	Audit/Event API

4. Docker Compose topology

The final compose topology contains 53 services and 8 named volumes. No duplicate host ports were detected during packaging validation. New healthchecks were added for v22.3 API services and new micro frontend containers.

Port	Container port	Service
2222	22	gitlab
3000	3000	grafana

Port	Container port	Service
3100	3100	loki
3200	3200	tempo
4200	80	frontend
4211	80	mfe-docker-desktop
4212	80	mfe-openlens-kubernetes
4213	80	mfe-task-management
4214	80	mfe-observability
4215	80	mfe-cicd-gitops
4216	80	mfe-terraform-studio
4217	80	mfe-devsecops
4218	80	mfe-ai-ops
4219	80	mfe-finops-cost
4220	80	mfe-infra-hub
4221	80	mfe-release-center
4222	80	mfe-policy-center
4223	80	mfe-notification-center
4317	4317	tempo
4318	4318	otel-collector
5672	5672	rabbitmq
6379	6379	redis
8080	8080	gateway-service
8081	8081	auth-service
8082	8082	task-service
8083	8083	notification-service
8084	8084	file-service
8085	8085	ai-ops-service
8086	8086	devsecops-service
8087	8087	pipeline-engine-service
8088	4180	mongo-express-sso
8089	4180	redis-commander-sso
8091	8091	go-cache-service
8092	8092	observability-service
8093	8093	gitops-control-service

5. Persistence and platform dependencies

Volume	Purpose
mongo-data	Named runtime persistence
redis-data	Named runtime persistence
rabbitmq-data	Named runtime persistence
grafana-data	Named runtime persistence
keycloak-db-data	Named runtime persistence
gitlab-config	Named runtime persistence
gitlab-logs	Named runtime persistence
gitlab-data	Named runtime persistence

6. Security and SSO

Keycloak remains the identity provider for shell, API services and supported tool access. The shell uses PKCE/OIDC; API services validate JWT/JWKS when Keycloak authorization is enabled; OAuth2 Proxy bridges continue to protect selected legacy tool UIs.